

NAME

TopologicalAtomTorsionsFingerprints

SYNOPSIS

```
use Fingerprints::TopologicalAtomTorsionsFingerprints;

use Fingerprints::TopologicalAtomTorsionsFingerprints qw(:all);
```

DESCRIPTION

TopologicalAtomTorsionsFingerprints class provides the following methods:

new, GenerateFingerprints, GetAtomTorsionsIDs, GetDescription, SetAtomIdentifierType, SetAtomicInvariantsToUse, SetFunctionalClassesToUse, StringifyTopologicalAtomTorsionsFingerprints

TopologicalAtomTorsionsFingerprints is derived from Fingerprints class which in turn is derived from ObjectProperty base class that provides methods not explicitly defined in TopologicalAtomTorsionsFingerprints, Fingerprints or ObjectProperty classes using Perl's AUTOLOAD functionality. These methods are generated on-the-fly for a specified object property:

```
Set<PropertyName>(<PropertyValue>);
$PropertyValue = Get<PropertyName>();
Delete<PropertyName>();
```

The current release of MayaChemTools supports generation of TopologicalAtomTorsionsFingerprints corresponding to following AtomIdentifierTypes:

```
AtomicInvariantsAtomTypes, DREIDINGAtomTypes, EStateAtomTypes,
FunctionalClassAtomTypes, MMFF94AtomTypes, SLogPAtomTypes,
SYBYLAtomTypes, TPSAAtomTypes, UFFAtomTypes
```

Based on the values specified for AtomIdentifierType along with other specified parameters such as AtomicInvariantsToUse and FunctionalClassesToUse, initial atom types are assigned to all non-hydrogen in a molecule. All unique atom torsions are identified and an atom torsion identifier is generated; the format of atom torsion identifier is:

```
<AtomType1>-<AtomType2>-<AtomType3>-<AtomType4>
```

AtomType1, AtomType2, AtomType3, AtomType4: Assigned atom types

where AtomType1 <= AtomType2 <= AtomType3 <= AtomType4

The atom torsion identifiers for all unique atom torsions corresponding to non-hydrogen atoms constitute topological atom torsions fingerprints of the molecule.

The current release of MayaChemTools generates the following types of topological atom torsions fingerprints vector strings:

```
FingerprintsVector;TopologicalAtomTorsions:AtomicInvariantsAtomTypes;3
3;NumericalValues;IDsAndValuesString;C.X1.B01.H3-C.X3.B03.H1-C.X3.B04-
C.X3.B04 C.X1.B01.H3-C.X3.B03.H1-C.X3.B04-N.X3.B03 C.X2.B02.H2-C.X2.B0
2.H2-C.X3.B03.H1-C.X2.B02.H2 C.X2.B02.H2-C.X2.B02.H2-C.X3.B03.H1-0...;
2 2 1 1 2 2 1 1 3 4 8 4 2 2 6 2 2 1 2 1 1 2 1 1 2 6 2 4 2 1 3 1
```

```
FingerprintsVector;TopologicalAtomTorsions:AtomicInvariantsAtomTypes;3
3;NumericalValues;IDsAndValuesPairsString;C.X1.B01.H3-C.X3.B03.H1-C.X3
.B04-C.X3.B04 2 C.X1.B01.H3-C.X3.B03.H1-C.X3.B04-N.X3.B03 2 C.X2.B02.H
2-C.X2.B02.H2-C.X3.B03.H1-C.X2.B02.H2 1 C.X2.B02.H2-C.X2.B02.H2-C.X3.B
03.H1-0.X1.B01.H1 1 C.X2.B02.H2-C.X2.B02.H2-N.X3.B03-C.X3.B04 2 C.X2.B
02.H2-C.X3.B03.H1-C.X2.B02.H2-C.X3.B03.H1 2 C.X2.B02.H2-C.X3.B03.H1...
```

```
FingerprintsVector;TopologicalAtomTorsions:DREIDINGAtomTypes;27;Numeri
calValues;IDsAndValuesString;C_2-C_3-C_3-C_3 C_2-C_3-C_3-0_3 C_2-C_R-C
_R-C_3 C_2-C_R-C_R-C_R C_2-C_R-C_R-N_R C_2-N_3-C_R-C_R C_3-C_3-C_2-0_2
C_3-C_3-C_2-0_3 C_3-C_3-C_3-C_3 C_3-C_3-C_3-N_R C_3-C_3-C_3-0_3 C...;
1 1 1 2 1 2 1 1 3 1 3 2 2 2 1 1 1 3 1 2 2 32 2 2 5 3 1
```

```
FingerprintsVector;TopologicalAtomTorsions:EStateAtomTypes;36;Numerical
lValues;IDsAndValuesString;aaCH-aaCH-aaCH-aaCH aaCH-aaCH-aaCH-aasC aaC
H-aaCH-aasC-aaCH aaCH-aaCH-aasC-aasC aaCH-aaCH-aasC-sF aaCH-aaCH-aasC-
ssNH aaCH-aasC-aasC-aasC aaCH-aasC-aasC-aasN aaCH-aasC-ssNH-dssC a...;
4 4 8 4 2 2 6 2 2 2 4 3 2 1 3 3 2 2 2 1 2 1 1 1 2 1 1 1 1 2 1 1 2
```

```
FingerprintsVector;TopologicalAtomTorsions:FunctionalClassAtomTypes;26
;NumericalValues;IDsAndValuesString;Ar-Ar-Ar-Ar Ar-Ar-Ar-Ar.HBA Ar-Ar-
Ar-HBD Ar-Ar-Ar-Hal Ar-Ar-Ar-None Ar-Ar-Ar.HBA-Ar Ar-Ar-Ar.HBA-None Ar
-Ar-HBD-None Ar-Ar-None-HBA Ar-Ar-None-HBD Ar-Ar-None-None Ar-Ar.H...;
32 5 2 2 3 3 3 2 2 2 2 1 2 1 1 1 2 1 1 1 1 3 1 1 1 3
```

```
FingerprintsVector;TopologicalAtomTorsions:MMFF94AtomTypes;43;Numerical
lValues;IDsAndValuesString;C5A-C5B-C5B-C5A C5A-C5B-C5B-C=ON C5A-C5B-C5
B-CB C5A-C5B-C=ON-NC=O C5A-C5B-C=ON-O=CN C5A-C5B-CB-CB C5A-CB-CB-CB C5
A-N5-C5A-C5B C5A-N5-C5A-CB C5A-N5-C5A-CR C5A-N5-CR-CR C5B-C5A-CB-C...;
1 1 1 1 1 2 2 2 1 1 2 2 2 2 1 1 2 1 1 2 1 1 1 2 1 1 1 2 18 2 2 1 1
1 1 2 1 1 3 1 3
```

```
FingerprintsVector;TopologicalAtomTorsions:SLogPAtomTypes;49;Numerical
Values;IDsAndValuesPairsString;C1-C10-N11-C20 1 C1-C10-N11-C21 1 C1-C1
1-C21-C21 2 C1-C11-C21-N11 2 C1-CS-C1-C10 1 C1-CS-C1-C5 1 C1-CS-C1-CS
2 C10-C1-CS-02 1 C10-N11-C20-C20 2 C10-N11-C21-C11 1 C10-N11-C21-C21 1
C11-C21-C21-C20 1 C11-C21-C21-C5 1 C11-C21-N11-C20 1 C14-C18-C18-C20
2 C18-C14-C18-C18 2 C18-C18-C14-F 2 C18-C18-C18-C18 4 C18-C18-C18-C...
```

```
FingerprintsVector;TopologicalAtomTorsions:SYBYLAtomTypes;26;Numerical
Values;IDsAndValuesPairsString;C.2-C.3-C.3-C.3 1 C.2-C.3-C.3-O.3 1 C.2
-C.ar-C.ar-C.3 1 C.2-C.ar-C.ar-C.ar 2 C.2-C.ar-C.ar-N.ar 1 C.2-N.am-C.
ar-C.ar 2 C.3-C.3-C.2-O.co2 2 C.3-C.3-C.3-C.3 3 C.3-C.3-C.3-N.ar 1 C.3
-C.3-C.3-O.3 3 C.3-C.3-C.ar-C.ar 2 C.3-C.3-C.ar-N.ar 2 C.3-C.3-N.ar-C.
ar 2 C.3-C.ar-C.ar-C.ar 1 C.3-C.ar-N.ar-C.3 1 C.3-C.ar-N.ar-C.ar 1 ...
```

```
FingerprintsVector;TopologicalAtomTorsions:TPSAAtomTypes;8;NumericalVa
lues;IDsAndValuesPairsString;N21-None-None-None 9 N7-None-None-None 4
None-N21-None-None 10 None-N7-None-None 3 None-N7-None-O3 1 None-None-
None-None 44 None-None-None-O3 3 None-None-None-O4 5
```

```
FingerprintsVector;TopologicalAtomTorsions:UFFAtomTypes;27;NumericalVa
lues;IDsAndValuesPairsString;C_2-C_3-C_3-C_3 1 C_2-C_3-C_3-O_3 1 C_2-C
_R-C_R-C_3 1 C_2-C_R-C_R-C_R 2 C_2-C_R-C_R-N_R 1 C_2-N_3-C_R-C_R 2 C_3
-C_3-C_2-O_2 1 C_3-C_3-C_2-O_3 1 C_3-C_3-C_3-C_3 3 C_3-C_3-C_3-N_R 1 C
_3-C_3-C_3-O_3 3 C_3-C_3-C_R-C_R 2 C_3-C_3-C_R-N_R 2 C_3-C_3-N_R-C_R 2
C_3-C_R-C_R-C_R 1 C_3-C_R-N_R-C_3 1 C_3-C_R-N_R-C_R 1 C_3-N_R-C_R-...
```

METHODS

new

```
$NewTopologicalAtomTorsionsFingerprints = new TopologicalAtomTorsionsFingerprints(
    %NamesAndValues);
```

Using specified *TopologicalAtomTorsionsFingerprints* property names and values hash, new method creates a new object and returns a reference to newly created *TopologicalAtomTorsionsFingerprints* object. By default, the following properties are initialized:

```
Molecule = ''
Type = 'TopologicalAtomTorsions'
AtomIdentifierType = ''
AtomicInvariantsToUse = ['AS', 'X', 'BO', 'H', 'FC']
FunctionalClassesToUse = ['HBD', 'HBA', 'PI', 'NI', 'Ar', 'Hal']
```

Examples:

```
$TopologicalAtomTorsionsFingerprints = new TopologicalAtomTorsionsFingerprints(
    'Molecule' => $Molecule,
    'AtomIdentifierType' =>
        'AtomicInvariantsAtomTypes');
```

```

$TopologicalAtomTorsionsFingerprints = new TopologicalAtomTorsionsFingerprints(
    'Molecule' => $Molecule,
    'AtomIdentifierType' =>
        'AtomicInvariantsAtomTypes',
    'AtomicInvariantsToUse' =>
        ['AS', 'X', 'BO', 'H', 'FC'] );

$TopologicalAtomTorsionsFingerprints = new TopologicalAtomTorsionsFingerprints(
    'Molecule' => $Molecule,
    'AtomIdentifierType' =>
        'DREIDINGAtomTypes');

$TopologicalAtomTorsionsFingerprints = new TopologicalAtomTorsionsFingerprints(
    'Molecule' => $Molecule,
    'AtomIdentifierType' =>
        'SYBYLAtomTypes');

$TopologicalAtomTorsionsFingerprints = new TopologicalAtomTorsionsFingerprints(
    'Molecule' => $Molecule,
    'AtomIdentifierType' =>
        'SLogPAtomTypes');

$TopologicalAtomTorsionsFingerprints = new TopologicalAtomTorsionsFingerprints(
    'Molecule' => $Molecule,
    'AtomIdentifierType' =>
        'FunctionalClassAtomTypes',
    'FunctionalClassesToUse' =>
        ['HBD', 'HBA', 'PI', 'NI', 'Ar', 'Hal']
);

$TopologicalAtomTorsionsFingerprints->GenerateFingerprints();
print "$TopologicalAtomTorsionsFingerprints\n";

```

GetDescription

```
$Description = $TopologicalAtomTorsionsFingerprints->GetDescription();
```

Returns a string containing description of topological atom torsions fingerprints.

GenerateFingerprints

```
$TopologicalAtomTorsionsFingerprints->GenerateFingerprints();
```

Generates topological atom torsions fingerprints and returns *TopologicalAtomTorsionsFingerprints*.

GetAtomTorsionsIDs

```
$AtomPairIDsRef = $TopologicalAtomTorsionsFingerprints->GetAtomTorsionsIDs();
@AtomPairIDs = $TopologicalAtomTorsionsFingerprints->GetAtomTorsionsIDs();
```

Returns atom torsion IDs corresponding to atom torsion count values in topological atom torsions fingerprints vector as an array or reference to an array.

SetAtomIdentifierType

```
$TopologicalAtomTorsionsFingerprints->SetAtomIdentifierType($IdentifierType);
```

Sets atom *IdentifierType* to use during atom torsions fingerprints generation and returns *TopologicalAtomTorsionsFingerprints*.

Possible values: *AtomicInvariantsAtomTypes*, *DREIDINGAtomTypes*, *EStateAtomTypes*, *FunctionalClassAtomTypes*, *MMFF94AtomTypes*, *SLogPAtomTypes*, *SYBYLAtomTypes*, *TPSAAAtomTypes*, *UFFAtomTypes*.

SetAtomicInvariantsToUse

```
$TopologicalAtomTorsionsFingerprints->SetAtomicInvariantsToUse($ValuesRef);
$TopologicalAtomTorsionsFingerprints->SetAtomicInvariantsToUse(@Values);
```

Sets atomic invariants to use during *AtomicInvariantsAtomTypes* value of *AtomIdentifierType* for topological atom torsions fingerprints generation and returns *TopologicalAtomTorsionsFingerprints*.

Possible values for atomic invariants are: AS, X, BO, LBO, SB, DB, TB, H, Ar, RA, FC, MN, SM. Default

value: AS,X,BO,H,FC.

The atomic invariants abbreviations correspond to:

AS = Atom symbol corresponding to element symbol

X<n> = Number of non-hydrogen atom neighbors or heavy atoms
 BO<n> = Sum of bond orders to non-hydrogen atom neighbors or heavy atoms
 LB0<n> = Largest bond order of non-hydrogen atom neighbors or heavy atoms
 SB<n> = Number of single bonds to non-hydrogen atom neighbors or heavy atoms
 DB<n> = Number of double bonds to non-hydrogen atom neighbors or heavy atoms
 TB<n> = Number of triple bonds to non-hydrogen atom neighbors or heavy atoms
 H<n> = Number of implicit and explicit hydrogens for atom
 Ar = Aromatic annotation indicating whether atom is aromatic
 RA = Ring atom annotation indicating whether atom is a ring
 FC<+n/-n> = Formal charge assigned to atom
 MN<n> = Mass number indicating isotope other than most abundant isotope
 SM<n> = Spin multiplicity of atom. Possible values: 1 (singlet), 2 (doublet) or 3 (triplet)

Atom type generated by AtomTypes::AtomicInvariantsAtomTypes class corresponds to:

AS.X<n>.BO<n>.LB0<n>.<SB><n>.<DB><n>.<TB><n>.H<n>.Ar.RA.FC<+n/-n>.MN<n>.SM<n>

Except for AS which is a required atomic invariant in atom types, all other atomic invariants are optional. Atom type specification doesn't include atomic invariants with zero or undefined values.

In addition to usage of abbreviations for specifying atomic invariants, the following descriptive words are also allowed:

X : NumOfNonHydrogenAtomNeighbors or NumOfHeavyAtomNeighbors
 BO : SumOfBondOrdersToNonHydrogenAtoms or SumOfBondOrdersToHeavyAtoms
 LB0 : LargestBondOrderToNonHydrogenAtoms or LargestBondOrderToHeavyAtoms
 SB : NumOfSingleBondsToNonHydrogenAtoms or NumOfSingleBondsToHeavyAtoms
 DB : NumOfDoubleBondsToNonHydrogenAtoms or NumOfDoubleBondsToHeavyAtoms
 TB : NumOfTripleBondsToNonHydrogenAtoms or NumOfTripleBondsToHeavyAtoms
 H : NumOfImplicitAndExplicitHydrogens
 Ar : Aromatic
 RA : RingAtom
 FC : FormalCharge
 MN : MassNumber
 SM : SpinMultiplicity

AtomTypes::AtomicInvariantsAtomTypes module is used to assign atomic invariant atom types.

SetFunctionalClassesToUse

```
$TopologicalTorsionsFingerprints->SetFunctionalClassesToUse($ValuesRef);
$TopologicalTorsionsFingerprints->SetFunctionalClassesToUse(@Values);
```

Sets functional classes invariants to use during *FunctionalClassAtomTypes* value of *AtomIdentifierType* for topological atom torsions fingerprints generation and returns *TopologicalAtomTorsionsFingerprints*.

Possible values for atom functional classes are: Ar, CA, H, HBA, HBD, Hal, NI, PI, RA. Default value [Ref 24]: HBD,HBA,PI,NI,Ar,Hal.

The functional class abbreviations correspond to:

HBD: HydrogenBondDonor
 HBA: HydrogenBondAcceptor
 PI : PositivelyIonizable
 NI : NegativelyIonizable
 Ar : Aromatic
 Hal : Halogen
 H : Hydrophobic
 RA : RingAtom
 CA : ChainAtom

Functional class atom type specification for an atom corresponds to:

Ar.CA.H.HBA.HBD.Hal.NI.PI.RA or None

AtomTypes::FunctionalClassAtomTypes module is used to assign functional class atom types. It uses following definitions [Ref 60-61, Ref 65-66]:

```
HydrogenBondDonor: NH, NH2, OH  
HydrogenBondAcceptor: N[!H], O  
PositivelyIonizable: +, NH2  
NegativelyIonizable: -, C(=O)OH, S(=O)OH, P(=O)OH
```

```
StringifyTopologicalAtomTorsionsFingerprints
```

```
$String = $TopologicalAtomTorsionsFingerprints->  
    StringifyTopologicalAtomTorsionsFingerprints();
```

Returns a string containing information about *TopologicalAtomTorsionsFingerprints* object.

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SEE ALSO

Fingerprints.pm, FingerprintsStringUtil.pm, AtomNeighborhoodsFingerprints.pm, AtomTypesFingerprints.pm, EStateIndicesFingerprints.pm, ExtendedConnectivityFingerprints.pm, MACCSKeys.pm, PathLengthFingerprints.pm, TopologicalAtomPairsFingerprints.pm, TopologicalAtomTripletsFingerprints.pm, TopologicalPharmacophoreAtomPairsFingerprints.pm, TopologicalPharmacophoreAtomTripletsFingerprints.pm

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